

Potential for Decarbonization in Bulk Transportation: Past, Present and Future

Emmanuel Coates

ECON128: Energy Economics & Policy

Professor Laila Safavi

May 8th, 2026

Introduction:

Long before a product reaches a shelf, it has already left a carbon footprint. The trucks that haul raw materials, the freight networks that distribute goods across continents, the logistics chains that keep modern supply systems running—each link burns fuel, emits carbon, and yet largely evades scrutiny. The bulk transportation sector accounts for a significant share of global greenhouse gas emissions (approximately $\frac{1}{3}$), and yet, they are never the target of regulatory policy (*Transportation Sector Emissions* | US EPA, 2025). Bulk ground transportation—the freight trucks and railroad networks that carry the physical weight of global commerce—operates in a policy gray zone: too economically vital to slow down, yet too emissions-intensive to be ignored. Understanding how this sector came to occupy that position, and what it would take to usher it into the era of energy efficiency is one of the more pressing and underexamined questions in climate policy today.

The story of decarbonization in bulk ground transportation is one of partial measures and deferred ambition. From early voluntary green practices to market based mechanisms like cap-and-trade and carbon offsets, each successive policy generation has moved us closer to the problem, without fully solving it. This has happened, not because the tools don't exist, but rather, because the frameworks governing this sector were never designed to withstand regulatory capture. With a governance architecture that prioritizes economic continuity over emissions accountability, accompanied by fragmented regulatory regimes, and an over-reliance on voluntary corporate

commitment, our emissions goals never stood a chance. Consequently, our regulatory policy has produced the appearance of progress, while emissions have continued largely unabated.

This paper follows that trajectory—examining not just how the bulk ground transportation sector has managed to evade meaningful decarbonization, but why the regulatory frameworks designed to address it have proven so persistently inadequate. It proceeds in three parts: first, tracing the historical development of fuel regulation and its structural limitations; second, assessing the effectiveness of current policy mechanisms such as carbon trading and cap-and-trade via case study; and finally, drawing on both to articulate what a credible path towards decarbonization in this sector must require.

The History of Fuel Regulation: From Railroad to Truck

The history of fuel regulation in the United States does not begin with emissions—it begins with economics. The first attempt at nationwide regulation of the transportation sector was the Interstate Commerce Act of 1887, which established the Interstate Commerce Commission (ICC) to manage the monopolistic pricing practices of the railroad industry. On the surface, this was a foundational consumer protection measure that would go on to define regulation for generations—a necessary act of exemplary punishment on an industry that had grown too powerful for laissez faire economics. Beneath the surface, however, the ICC's impact would suggest a more

nuanced relationship, one in which an agency designed specifically to constrain the ground transportation industry was ultimately reshaped by it.

The ICC's drift towards regulatory capture was neither immediate nor accidental. Once railroad companies recognized that federal oversight was inevitable, they moved strategically to influence its terms; lobbying for commissioners who would be sympathetic to industry concerns, narrowing the scope of enforceable provisions, and over time, converting the regulatory apparatus from a constraint into a shield against new competition. It is important to note that this was not accomplished by the acts of one firm, this was made possible because of the unique commonality of interest between shippers, railroad companies, and the public (Thompson, 2000). Shippers wanted stable rates, railroad companies saw an opportunity to limit competition, while the public—as a result of information asymmetry, or nescience, or a combination of the two—seemed content that they had stuck it to the man. While this may not constitute corruption in the conventional sense, it reveals something more enduring: a regulatory body that had internalized the priorities of the industry that it was meant to oversee.

To understand why regulatory capture proved so durable, one must first recognize that the interests of the corporation and the state were never truly disjoint. Economic growth was their nexus; profitable industries meant more tax revenue, more employment, and more political capital. Regulation was never meant to disrupt that balance, but to enhance it. So when environmental regulation finally arrived in the mid-twentieth century, it didn't displace the economic logic that had been governing the

transportation industry, instead it coalesced with it, inheriting the same structural vulnerabilities as its predecessor.

With the development of the automobile—along with its widespread popularity and adoption—in the mid-twentieth century, came our first piece of federal legislation aimed at controlling air pollution, the Clean Air Act of 1963. Yet from its inception, the Act bore the marks of the same negotiated compromise that had defined transportation regulation since 1887—provocative, but perfunctory. It funded research, encouraged state action, and stopped well short of mandating anything—not even an emissions standard (*Evolution of the Clean Air Act* | US EPA, 2015).

The Clean Air Act Amendments of 1970 marked a genuine inflection point in the history of U.S. environmental regulation. Where the 1963 Act had pointed out the problem, the 1970 Amendments confronted it; establishing, for the first time, binding national ambient air quality standards and requiring automakers to equip almost all vehicles with catalytic converters (*Dujack*, 2024). And it didn't stop there, the act also expanded federal authority to cover both stationary sources, such as industrial facilities, in addition to mobile sources like automobiles and trucks, which had become ubiquitous. In parallel, the National Environmental Policy Act established the Environmental Protection Agency to implement the provisions outlined by the Clean Air Act of 1970 (Ross, 2012). While it took Congress another seven years, a burgeoning environmental movement, and the political pressure of a nation choking on smog, they did, in effect, accept some degree of accountability in acknowledging clean air as a public good worth protecting.

This momentum, however, would not go unchallenged. In 1973, the Organization of Arab Petroleum Exporting countries announced that they would be implementing an embargo on countries that had supported Israel during the Yom Kippur War, forcing the fuel economy back onto the legislative agenda (*Milestones in the History of U.S. Foreign Relations - Office of the Historian*, 2026). As a result, Congress passed the Corporate Average Fuel Economy (CAFE) standards of 1975, aimed at reducing our dependence on fuel imports by requiring automakers to make their fleets more fuel efficient. While this policy was inherently green—and is still considered so—that was never its intent, and if it was then Congress severely missed the mark. While they accomplished their goal of making cars more fuel efficient, and thereby reducing fuel consumption, the reductions that were initially observed were phantoms (*CAFE Standards - R Street Institute*, 2025). What appeared to be a reduction in consumption was quickly offset by the rebound effect, because as vehicles became cheaper to run, Americans simply drove more, rendering it unsuccessful in the aggregate.

CAFE standards did not fail to reduce fuel consumption because of a counterproductive populace, but rather because our regulation is so thoroughly shaped by industry priorities, that even when they meet their desired purpose, corporations are there to absorb the benefits. Nowhere is that more evident than in the treatment of light trucks and SUVs under the original CAFE framework. When the standards were enacted in 1975, light trucks—a category that would soon come to encompass the SUV—were subject to a significantly lower fuel efficiency requirement than passenger cars, on the grounds that they primarily served commercial and agricultural purposes. What

followed was one of the most consequential acts of regulatory arbitrage in American automotive history: automakers recognized that the light truck classification offered a profitable escape from the stricter standard imposed on passenger vehicles, and systematically shifted their production. As of 2023, SUVs dominate the roads and account for 48% of car sales, a direct consequence of a loophole that was realized too late (IEA, 2024).

This rich history of emissions regulatory insignificance reveals not a series of isolated policy failures, but rather a pattern wherein corporations are able to convince regulatory bodies that what benefits them benefits us all. From the ICC's capture by railroad interests in the nineteenth century, to the phantom emissions reductions, and loopholes engendered by CAFE, American transportation regulation has remained nothing if not consistent: ambitious in declaration, negotiated into inadequacy, and ultimately absorbed by the industries it was designed to constrain. This isn't a result of incompetence, nor is it coincidence—it's the structural codependence of the state and the corporation, bound together by a shared investment in economic growth that has persistently outweighed any commitment to environmental accountability. Each of these moments reveals that it wasn't regulation that failed to be effective, but rather the regulator—decarbonization in the bulk transportation sector hasn't worked because it's always been debatable, and what remains to be seen is whether our new mechanisms follow this trajectory, or deviate from it.

Current Mechanisms, Persistent Gaps: A Case Study in Corporate Decarbonization: CHEP and the limits of Voluntary Commitment

The question posed at the conclusion of the previous section—whether or not our new mechanisms have deviated from their pattern of compromise—demands an honest discussion of what those mechanisms are, their intended purpose, and the outcome that they have delivered. To understand how the failures of current policy mechanisms manifest at the level of individual corporate actors, this section examines one of the largest bulk logistics operations in the world, the Commonwealth Handling Equipment Pool (CHEP), a subsidiary of Brambles Company. CHEP is often considered an industry leader in sustainability; they hold Science-Based Targets validated by the Science-Based Targets initiative (SBTi), have publicly committed to a net-zero emissions standard by 2040, and claim that their logistics operations have been carbon-neutral since 2021 (Brambles, 2025). It's precisely this reputation that makes CHEP worth examining in more detail; because underneath the mask of corporate social responsibility lies not a green corporate actor constrained by poor policy, but a profit-maximizing monopoly that has used their dominant market position to eliminate competition, capture regulatory favor, and perform, rather than practice its commitment to environmental accountability.

The first sign of a ruse, their claim to carbon neutrality. Throughout the previous sections we have established that the transport of goods is incredibly emissions intensive because they have to travel long distances, often by trucks or rail, both of

which emit GHG. Furthermore, this paper has traced the history of regulation on the bulk transportation industry, which would suggest the exact opposite of what CHEP is claiming. So what's missing? How is CHEP claiming carbon neutrality? The answer is simple: they're buying it. The Brambles' FY25 Annual Report reveals that carbon neutrality for Scope 1 and 2 operations—which cover direct, and indirect emissions, from purchased energy—were achieved not through emissions elimination but through the purchase of voluntary carbon offsets and Energy Attribute Certificates (Brambles, 2025).

The mechanism CHEP relies upon to achieve its carbon neutral status is not merely imperfect, but it's systematically fraudulent. A 2024 meta analysis examining nearly one billion tons of credits found that less than one-sixth of the projects included represented any meaningful reductions (Romm, 2025). How does this happen? Carbon offsets depend on counterfactual baselines—projections of what emissions would have been without the offset program—but the problem is that these baselines are constructed by the very developers seeking to profit from the credits, creating an information asymmetry that is nearly impossible for regulators to correct (Romm, 2025). Validators aren't regulators themselves, but rather third party companies who depend on demand from companies like CHEP who seek their services. In this sense, overstating emissions reductions through weak validations become a unique advantage to these firms, and consistent with Gresham's law, as the demand for bad credits drives out the good ones.

This brings us back to corporate strategy, companies like CHEP don't just stumble into weak validators; they seek them out, because in systems lacking proper

enforcement mechanisms, the cheapest credit is the best credit. Consider it this way, the reputational reward for a claim of carbon neutrality is substantial, and immediately marketable, while the consequences of overcrediting are delayed, and furthermore, borne by the atmosphere instead of the balance sheet. It's no coincidence that the registries CHEP has selected to validate its carbon neutrality—Verra and Gold Standard—are the same programs responsible for overcrediting (Brambles, 2025; Romm, 2025). You can trust that this isn't a mistake, but the most efficient way of purporting corporate social responsibility without actually being responsible.

The carbon neutrality claim, however, only covers the emissions from Scope 1 and 2; the smallest slice of CHEP's emissions footprint. The more significant finding lies in Scope 3: downstream transportation and distribution alone accounts for 697,600 tons of CO₂ emissions in FY25, which is more than half of Brambles' total emissions—while the carbon neutrality claim only covers the 29,500 tonnes included in Scope 1 and 2 (Brambles, 2025). This means that CHEP is claiming carbon neutrality while covering less than 4% of their total emissions output. This isn't an oversight—this is what happens when you try to fix a problem while simultaneously ignoring the root cause. This is exactly the question that (Peng, 2024) identifies as a central failure of current policy: the freight operators and logistics subcontractors that physically move the goods from origin to destination, aren't being regulated by any mandatory downstream emissions trading system. So it seems surprising that companies like CHEP, in the bulk transportation sector, can claim carbon neutrality, because the actual emitters aren't facing regulations, and furthermore, they are not responsible for offsetting their emissions—the emissions that come directly out of their services. Brambles even

acknowledges this flaw, noting that they are actively “advocating for a supportive policy environment” (Brambles, 2025) in ground transport, yet, they don’t even cover the emissions that they are actually responsible for. With approximately 60% of its SBT-covered emissions coming from subcontracted services, CHEP’s decarbonization plan isn’t really a plan, it’s a conditional; deferred to a future the company has every strategic interest in delaying.

That strategic interest becomes clearer when you examine how CHEP operates within its own market. CHEP holds a dominant position in the pallet pooling industry, securing major wholesalers like Costco and Walmart through long-term contractual arrangements at predatorially low rental rates; rates calibrated not to reflect true cost, but to make competition virtually impossible for smaller operators. This isn't incidental to their business model, it is their business model. By charging artificially low rates, and evading responsibility for their high emissions, CHEP has captured volume, locking in customers through restrictive contracts, decimating the independent pallet recyclers and whitewood operators that would otherwise compete with them.

The efficiency gains CHEP offers to its customers are real, they’re just subsidized by the externalization of costs onto the smaller actors in its supply chain. Nowhere has this been more visible than in CHEP’s treatment of independent pallet recyclers in California. When CHEP’s pallets accumulate at recycler facilities faster than CHEP retrieves them, small operators are forced to absorb the storage and handling costs. Rather than compensate these recyclers fairly, CHEP deploys its Asset Recovery Program, retrieving their lost pallets back at rates its own representatives acknowledged

were not designed to be profitable for recyclers, and pursue legal action against companies acquiring its pallets—in good faith—in the ordinary course of business (California Assembly Committee on Privacy and Consumer Protection, 2022). California AB1760 was passed in 2022 precisely to correct the legal imbalance CHEP has exploited—and CHEP responded by challenging the constitutionality of the law in state court, filing separate lawsuits against two small pallet recyclers in Southern California, seeking three times their damages from recyclers it has spent years undercompensating (Pallet Enterprise, 2024) . Meanwhile, the state of Florida exempts CHEP pallets from sales tax under Fla. Admin Code Ann. R. 12a-1.040—a preferential regulatory treatment justified on the grounds of CHEP’s environmental contribution to the supply chain (Johnson, 1999). CHEP is clearly a company that has manipulated both the language of sustainability and the mechanisms of regulation, while litigating against the small competitors it is simultaneously undercutting. They're not a green corporate actor, they're the railroad company—fast-forward to the twenty-first century.

Conclusion: The Path Forward

The central failure documented throughout this paper is not technical, but rather architectural. Decarbonization in the Bulk ground transportation sector has not worked because regulation has never accurately addressed the source of the problem, and because the frameworks designed to constrain corporate emissions have instead been leveraged by corporations as instruments of competitive advantage and regulatory shelter. From the ICC to CAFE, to the voluntary offset market, the pattern is consistent:

ambitious in declaration, negotiated into inadequacy, and the benefits? Ultimately absorbed by the industries it was designed to constrain. A credible path forward begins not with another market mechanism layered atop a broken foundation, but with a fundamental reorientation—holding corporations directly accountable for the emissions they generate, and responsible for the abatement those emissions require.

That reorientation demands a mandatory downstream emissions trading system that places enforceable carbon obligations on the freight operators, logistics companies, and bulk transportation firms that constitute the actual engine of ground transportation emissions—not just the fuel distributors that precede them in the supply chain. Upstream interventions have failed because the carbon signal must travel through a cascade of intermediaries before it reaches the actual emitter (Peng, 2024). A downstream system bypasses that cascade entirely, assigning obligations directly to the entities burning the fuel, with a monitoring reporting and verification architecture robust enough to close the information asymmetries that have undermined every previous mechanism (Peng, 2024). Successful market based regulation requires that incentives are properly aligned with the behavior it is meant to change, but for three decades, transportation emissions policy has focused on the fuel alone, and not the act of burning it (Griffin & Puller, 2005).

What this also requires is a stronger, enforceable standard of carbon neutrality—not a corporate benchmark, not a weak certification purchased from a lenient registry, but a genuine accounting of emissions produced, and emissions eliminated. As Romm, Lezak and Alshamsi make clear, the voluntary carbon market in its current form does

not deliver emissions reductions; it delivers the paperwork that says it has (Romm, 2025). A credible standard would require direct emissions reductions as the primary obligation, with offsets permitted only as a residual instrument subject to rigorous, independently enforced verification. The goal this paper has been reaching toward—meaningful decarbonization in the bulk ground transportation sector—is achievable. But it won't be achieved by the same regulatory architecture that has deferred it for over a century. It will require regulation that is designed to deliver emission reductions, not corporate expansion.

References

- Alger, J., Lister, J., & Dauvergne, P. (2021). Corporate governance and the environmental politics of shipping. *Global Governance: A Review of Multilateralism and International Organizations*, 27(1), 144–166.
<https://doi.org/10.1163/19426720-02701001>
- Brambles Limited. (2025). *Brambles annual report FY2025*.
<https://brambles.com/ar2025>
- CAFE Standards. (2025, September 11). R Street Institute.
<https://www.rstreet.org/research/cafe-standards/>
- California Assembly Committee on Privacy and Consumer Protection. (2022). *AB 1760 bill analysis: Bulk merchandise pallets*.
<https://trackbill.com/s3/bills/CA/2021/AB/1760/analyses/assembly-privacy-and-consumer-protection.pdf>
- CHEP. (n.d.). *Carbon neutral platforms*. Retrieved May 5, 2026, from
<https://www.chep.com/files/download/CHEP%20Fresh%20Carbon%20Neutral%20Platforms.pdf>
- Dujack, S. (2024). *Earlier laws had lessons for Clean Air Act*. Environmental Law Institute. <https://www.eli.org/vibrant-environment-blog/earlier-laws-had-lessons-clean-air-act>
- Evolution of the Clean Air Act. (2015, May 29). U.S. Environmental Protection Agency.
<https://www.epa.gov/clean-air-act-overview/evolution-clean-air-act>
- Griffin, J. M., & Puller, S. L. (2005). Introduction: A primer on electricity and the economics of deregulation. In J. M. Griffin & S. L. Puller (Eds.), *Electricity deregulation: Choices and challenges* (pp. 1–28). University of Chicago Press.
<https://doi.org/10.7208/9780226308586-001>
- IEA. (2024). *SUVs are setting new sales records each year — and so are their emissions*. <https://www.iea.org/commentaries/suvs-are-setting-new-sales-records-each-year-and-so-are-their-emissions>

- Johnson, J. (1999, September 9). Florida offers pallet tax break. *Transport Topics*. <https://www.ttnews.com/articles/florida-offers-pallet-tax-break>
- Lai, K.-H., Lun, V. Y. H., Wong, C. W. Y., & Cheng, T. C. E. (2011). Green shipping practices in the shipping industry: Conceptualization, adoption, and implications. *Resources, Conservation and Recycling*, 55(6), 631–638. <https://doi.org/10.1016/j.resconrec.2010.12.004>
- Legislative Analyst's Office. (2025, May 7). *Assessing California's climate policies: Cap-and-trade reauthorization*. <https://lao.ca.gov/reports/2025/5042/Assessing-Californias-Climate-Policies-Cap-and-Trade-Reauthorization-050725.pdf>
- Milestones in the history of U.S. foreign relations: The 1973–74 oil embargo. (2026). Office of the Historian, U.S. Department of State. <https://history.state.gov/milestones/1969-1976/oil-embargo>
- Pallet Enterprise. (2024, October 1). *Recyclers and CHEP clash in California state court over fair compensation, conversion*. <https://palletenterprise.com/recyclers-and-chep-clash-in-california-state-court-over-fair-compensation-conversion/>
- Peng, H., Sun, Y., Hao, J., An, C., & Lyu, L. (2024). Carbon emissions trading in ground transportation: Status quo, policy analysis, and outlook. *Transportation Research Part D: Transport and Environment*, 131, 104225. <https://doi.org/10.1016/j.trd.2024.104225>
- Rodriguez, F. (2022). *Assembly Bill 1760: Bulk merchandise pallets*. California State Legislature. https://leginfo.ca.gov/faces/billNavClient.xhtml?bill_id=202120220AB1760
- Romm, J., Lezak, S., & Alshamsi, A. (2025). Are carbon offsets fixable? *Annual Review of Environment and Resources*, 50(1), 649–680. <https://doi.org/10.1146/annurev-environ-112823-064813>
- Ross, K., Chmiel, J. F., & Ferkol, T. (2012). The impact of the Clean Air Act. *The Journal of Pediatrics*, 161(5), 781–786. <https://doi.org/10.1016/j.jpeds.2012.06.064>
- Thompson, L. (n.d.). *Regulatory developments in the U.S.: History and philosophy*. Retrieved May 5, 2026, from https://tgaassoc.com/documents/index-109-REG_NOTE.PDF

Transportation sector emissions. (2025, January 9). U.S. Environmental Protection Agency. <https://www.epa.gov/ghgemissions/transportation-sector-emissions>